Matrix
Addition
Subtraction
Multiplication

What is a matrix ?

It is a grid of numbers

In a 2D matrix, you have rows and columns

The shape of matrix is **rows x columns**
Example: 2 x 3

$$\begin{array}{ccc} \text{Col.1} & \text{Col.2} & \text{Col.3} \\ \end{array}$$
$$\begin{bmatrix} 2 & 5 & 1 \\ 1 & 3 & 7 \end{bmatrix} \begin{array}{l} \rightarrow \text{Row 1} \\ \rightarrow \text{Row 2} \end{array}$$

Matrices are used to:

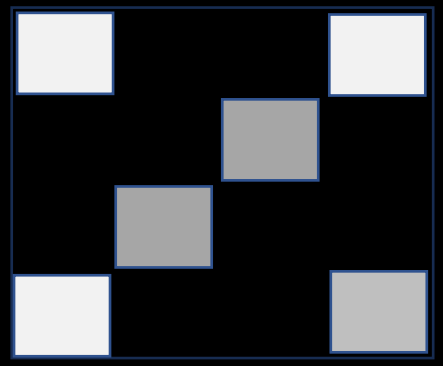
- Represent data (e.g., a dataset where rows = samples, columns = features).

age	income	experience
23	45000	6
61	65000	4
25	33000	1
51	78000	6

feature 1	feature 2	feature 3	
23	45000	6	→ data sample 1
61	65000	4	→ data sample 2
25	33000	1	
51	78000	6	

Matrices are used to images:

0 -> black
255 -> white



Input 4X4 image

203	0	0	200
0	0	30	0
0	32	0	0
205	0	0	49

Pixel representation in 2D (not normalized), Size = 4X4

matrix addition + multiplication

Matrix addition / subtraction:

The shape of 2 matrix should be same for addition or subtraction.

$$\underset{2\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}} + \underset{2\times 2}{\begin{bmatrix} 1 & 7 \\ 9 & 6 \end{bmatrix}} = \begin{bmatrix} 2+1 & 3+7 \\ 1+9 & -5+6 \end{bmatrix} = \begin{bmatrix} 3 & 10 \\ 10 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 1 & 7 \end{bmatrix} \Rightarrow \underline{\underline{\times}} \text{ not Possible}$$

Matrix subtraction:

$$\underset{2\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \end{bmatrix}} - \underset{2\times 2}{\begin{bmatrix} 1 & 7 \\ 9 & 6 \end{bmatrix}} = \begin{bmatrix} 2-1 & 3-7 \\ 1-9 & -5-6 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ -8 & -11 \end{bmatrix}$$

Matrix multiplication:

Rule:

$$[A]\ [B] = [C]$$

$m \times n \qquad n \times p \qquad m \times p$

must be same

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ -1 & 3 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 2$

$\Rightarrow$ matrix multiplication possible

C would be 2×2

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 5 & 2 \\ 3 & 7 & 9 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 3$

$\Rightarrow$ matrix multiplication possible

C would be 2×3

Matrix multiplication:

Rule:

$$[A]\ [B] = [C]$$

$m \times n \qquad n \times p \qquad m \times p$

must be same

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 9 \\ 5 & 6 \end{bmatrix} \Rightarrow$$

matrix multiplication NOT possible

$2 \times 2 \qquad 3 \times 2$

NOT same

Matrix multiplication:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

2×2 $\quad$ 2×2 $\quad$ 2×2

$c_{11} \rightarrow$ (Row 1 of A) $\cdot$ (Col. 1 of B)

$c_{12} \rightarrow$ (Row 1 of A) $\cdot$ (Col. 2 of B)

$c_{21} \rightarrow$ (Row 2 of A) $\cdot$ (Col. 1 of B)

$c_{22} \rightarrow$ (Row 2 of A) $\cdot$ (Col. 2 of B)

Matrix multiplication:

Suppose we have matrix A and B.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

We want C = AB

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} (1,2)\cdot(5,7) & (1,2)\cdot(6,8) \\ (3,4)\cdot(5,7) & (3,4)\cdot(6,8) \end{bmatrix}$$

$$= \begin{bmatrix} 1\times5 + 2\times7 & 1\times6 + 2\times8 \\ 3\times5 + 4\times7 & 3\times6 + 4\times8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Matrix multiplication:

Rule:

$$[A] \quad [B] = [C]$$

$$m \times n \qquad n \times p \qquad m \times p$$

must be same

The general formula for C:

$$C_{ij} = \sum_k A_{ik} B_{kj}$$

Meaning:
take row **i** from A
take column **j** from B
multiply matching elements(i.e. dot product)
add them

$$\begin{bmatrix} c_{11} & \cdots & \vdots \\ \vdots & \cdots & c_{ij} \cdots \\ & & \vdots \end{bmatrix}$$

Matrix multiplication:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 3 \end{bmatrix} = \begin{array}{c} \begin{array}{ccc} c_{11} & c_{12} & c_{13} \end{array} \\ \begin{bmatrix} (1,2)\cdot(5,8) & (1,2)\cdot(6,9) & (1,2)\cdot(7,3) \\ (3,4)\cdot(5,8) & (3,4)\cdot(6,9) & (3,4)\cdot(7,3) \end{bmatrix} \\ \begin{array}{ccc} c_{21} & c_{22} & c_{23} \end{array} \end{array}$$

2×2 $\quad$ 2×3 $\quad$ 2×3

Matrix multiplication are used to perform linear transformations: Scaling, rotation of images

Scaling Matrix:

Suppose you have matrix in 2D

$$S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \rightarrow \text{scaling matrix}$$

$s_x \rightarrow$ scale in x-direction

$s_y \rightarrow$ scale in y-direction

Matrices are used to Perform linear transformations: Scaling, rotation of images

Let $S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ & $P = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Apply scaling:

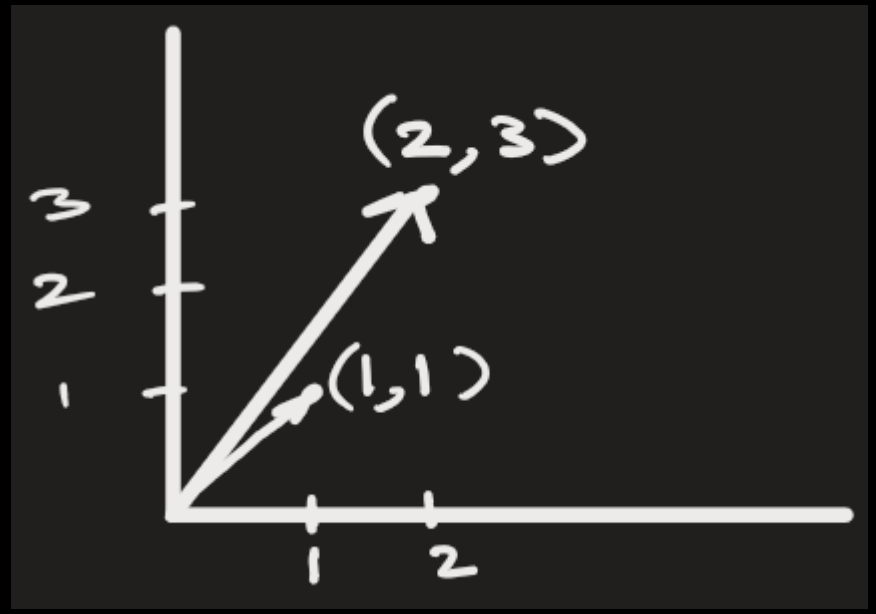
$$S \cdot P = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\times1 + 0\times1 \\ 0\times1 + 3\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

So, the point (1,1) becomes (2,3)

→ stretched along x-axis by 2

→ stretched along y-axis by 3



Rotation Matrix:

It rotates the vector.

It does not change the magnitude of the vector.

Rotation in 2D

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$\theta \rightarrow$ rotation angle

Rotation Matrix:

Rotate by 90°

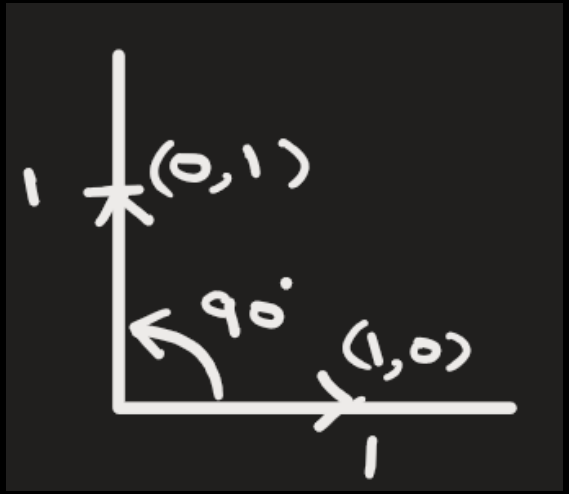
$$R(90^\circ) = \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Let $P = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$R \cdot P = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$P = (1,0)$ after rotation $\rightarrow (0,1)$



Combine Scaling + Rotation:

we can combine scaling + rotation:

$$T = R \cdot S$$

Let

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Then

$$T = R \cdot S = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

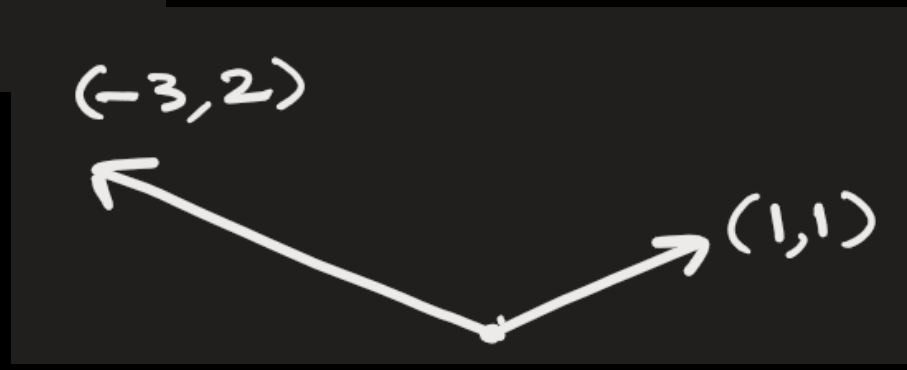
$$= \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix}$$

Apply T to a point $P = (1,1)$

$$P' = T \cdot P = \begin{bmatrix} 0 & -3 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

So,

$$P = (1,1) \xrightarrow{T} P' = (-3,2)$$



Uses of matrix multiplication:

Matrices transform coordinates of points in images.

- In graphics or robotics, when you perform scaling, rotating, or transforming an image, you are doing matrix multiplication.
- In ML when you perform scaling or transforming data , you are doing matrix multiplication.
- When you rotate or zoom an image → matrix multiplication is happening!

Representing an image as a matrix and then blurring it:

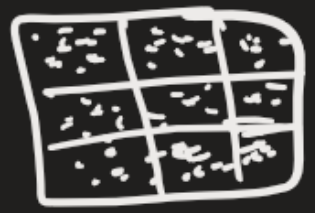
Suppose we have a grayscale image of size 3x3 pixels:

255, 0, 0
0, 255, 0
0, 0, 255

Each value represents brightness of a pixel (0=black, 255=white).

Applying a matrix operation — Blurring
Let’s apply a simple average blur filter (kernel):
1/9, 1/9, 1/9
1/9, 1/9, 1/9
1/9, 1/9, 1/9

$$\begin{bmatrix} 255, & 0, & 0 \\ 0, & 255, & 0 \\ 0, & 0, & 255 \end{bmatrix} \begin{bmatrix} 1/9, & 1/9, & 1/9 \\ 1/9, & 1/9, & 1/9 \\ 1/9, & 1/9, & 1/9 \end{bmatrix} = \begin{bmatrix} 28.3 & 28.3 & 28.3 \\ 28.3 & 28.3 & 28.3 \\ 28.3 & 28.3 & 28.3 \end{bmatrix}$$



This is just matrix multiplication

STOP

1) Predicting house prices using matrix operations for linear regression (see jupyter n/b) :

Predict house prices using:
y_hat = W*X
where,
X = feature matrix (house size, number of rooms,etc)
W = weight matrix (parameters to learn)
Y_hat = predicted price

Size (sqft)	Rooms	Price (target)
1000	3	300,000
1500	4	400,000
2000	3	500,000
2500	5	600,000
3000	4	700,000

✅ Matrix form

We add a column of 1 to handle the intercept (bias):

$$X = \begin{bmatrix} 1 & 1000 & 3 \\ 1 & 1500 & 4 \\ 1 & 2000 & 3 \\ 1 & 2500 & 5 \\ 1 & 3000 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 300{,}000 \\ 400{,}000 \\ 500{,}000 \\ 600{,}000 \\ 700{,}000 \end{bmatrix}$$

We want to find W so that:

$$\hat{y} = XW$$

W is determined using following mathematical formula:

$$y = X W$$
$$\Rightarrow W = (X^T \cdot X)^{-1} \cdot X^T \cdot y$$

$$W = \begin{bmatrix} 10^5 \\ 200 \\ -5.8 \times 10^{-11} \end{bmatrix}$$

Let say a new house has size = 1800 sqft
room = 4

then $\hat{y} = [1 \quad 1800 \quad 4] \begin{bmatrix} 10^5 \\ 200 \\ -5.8 \times 10^{-11} \end{bmatrix}$

$= \$460{,}000$

🔵 2. Types of Matrix Operations

We'll cover the main ones used in ML:

1. Addition
2. Scalar multiplication
3. Matrix multiplication
4. Transpose
5. Identity matrix
6. Inverse (conceptual)

- Explain how to find determinant

```
A = np.array([[2, 1],
              [3, 5]])

det_A = 2*5 - 1*3 = 7

A = np.array([[2, 1, 0],
              [0, 1, -1],
              [0, 2, 4]])
det_A = 12
```